Chapter 11 Answers

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Note : Smullyan must have made a typo when defining Σ_0 formulae, as he said that the existential type is $\exists x(x \leq \overline{n} \supset F(x))$, which makes no sense, as this is a valid because there is always some x that is greater than n, thus making the statement true. It seem that what he meant to say was $\exists x(x \leq \overline{n} \wedge F(x))$, and I will think of Σ_0 formulas as such. (I later checked pp. 94 of the Begginer's Further Guide, and his typo was confirmed, as he defines the existential case with a $\wedge$, not a $\supset$).

Problem 1

(a):
$x\ div\ y$ can be represented by the formula:
$\exists z(z \times x = y)$
(b):
$x\ pdiv\ y$ can be represented by the formula:
$x\ div\ y \wedge (\sim(x = 1) \wedge \sim(x = y))$
(c):
$x < y$ can be represente by the formula:
$\exists z(\sim(z = 0) \wedge z + x = y)$
(d):
$prm\ x$ can be represented by the formula:
$\sim\exists z(z\ pdiv\ x)$

Problem 2

Consider any set or relation that is expressible by some formula the includes $\leq$. Well, we can define $x \leq y$ as: $\exists z(z + x = y)$, thus meaning the same formula is expressible in terms of plus and minus.

Problem 3

We are being asked to prove that $y > 0$ and $x = 2^{L(y)}$ are true iff the following three conditions hold:
$C_1 : x - 1 \leq y \leq 2 \times x - 1$
$C_2 : x$ is a power of 2
$C_3 : y \neq 0$

Proof. On the one hand, assume that the three conditions C_1, C_2, C_3 all hold. Well, by C_3, $y \neq 0$. Hence, because y must be a positive integer, it follows that $y > 0$. Next, we know by C_2 that x is a power of 2, and thus, $x = 2^m$ for some m. Hence, substituting into C_1, we get:
$2^m - 1 \leq y \leq 2 \times 2^m - 1$
Then, by $(*)$, we know that $m = r$ for $r = L(y)$. Hence, we know that $x = 2^m = 2^r = 2^{L(y)}$, thus completing this case of the proof.

On the other hand, assume that $y > 0$ and $x = 2^{L(y)}$. Hence, C_3 holds, as $y \neq 0$. Also, because $x = 2^{L(y)}$, C_2 holds, as x is a power of 2. Lastly, because $x = 2^{L(y)}$ where $L(y) = r$ for r the length of y, we know $x = 2^r$, and thus we can substitute, yielding $x - 1 \leq y \leq 2 \times (x - 1)$ by $(*)$. □

Problem 4

C_1 is true iff:
$(\exists z \leq x)(z = x + 1 \wedge z \leq y \wedge y \leq 2 \times z)$

Problem 5

(a):
$Pow_2(x)$ can be expressed by the formula:
$(\forall y \leq x)((y \; div \; x \wedge y \neq 1) \supset 2 \; div \; x)$
(b):
First, $x \neq y$ can be expressed by the formula $\sim(x = y)$. So, we can represent the desired relation with the formula:
$x \neq 0 \wedge y \neq 0 \wedge (\exists v \leq z)(v = 2^{L(y)} \wedge z = x \times v + y)$
by Problem 3. We can tell this is Σ_0, as, because all of C_1, C_2, C_3 are Σ_0, we have their conjunction $v = 2^{L(y)}$ also being Σ_0.

Problem 6

(a):
Well, because the concatenation operation is assosciative, $x * y * z = (x * y) * z$. Thus, by Problem 5(b), $(x*y)*z$ is Σ_0. Hence, the relation under consideration can be represented by the Σ_0 formula:
$0 \neq x_1 \wedge 0 \neq x_2 \wedge 0 \neq x_3 \wedge (\exists v \leq z)(v = x_1 * x_2 \wedge z = v * x_3)$
(b):
The base case was proven in Problem 5(a). Now, make an induction assumption. Hence:
$0 \neq x_1 \wedge ... \wedge 0 \neq x_n \wedge x_1 * x_2 * ... * x_n = y$ is Σ_0. In the case for $n+1$, we have the following Σ_0 formula representing the desired relation by the induction assumption, as well as Problem 5(a): $0 \neq x_1 \wedge ... \wedge 0 \neq x_n + 1 \wedge (\exists z \leq y)(z = x_1 * x_2 * ... * x_n \wedge y = z * x_n + 1)$

Problem 7

(a):
$x\ B\ y$ can be represented by the Σ_0 formula:
$(\exists z \leq y)(xz = y \vee x = y) \wedge (x \neq 0 \wedge y \neq 0)$ (b):
$x\ E\ y$ can be represented by the Σ_0 formula:
$(\exists z \leq y)(zx = y \vee x = y) \wedge (x \neq 0 \wedge y \neq 0)$ (c):
$x\ P\ y$ can be represented by the Σ_0 formula:
$(x\ B\ y \vee x\ E\ y) \vee (\exists z_1 \leq y)(\exists z_2 \leq y)(z_1 x z_2 = y)$

Exercise

(A question I had great confusion over!)
We are considering some number z satisfying the following two conditions:
(1): z contains at least one frame.
(2): If f is the largerst frame in z, then there is at least one string in z of the form $ffafbff$ where f is not a part of a or b.
We are to show that this number z encodes one and only one sequence of positive integers.

Proof. Let us recall the definition of what it means for a finite sequence S of ordered pairs of positive integers to be encoded in some z (on pp. 205):
C_1: The longest frame f in z contains a string of 1's longer than any string of 1's in any of the numbers in S.
C_2: Every ordered pair (a, b) in S is found within z in a string of the form

$ffafbff$, where f occurs in neither a nor b, and with f being the longest frame occuring in z.
C_3: The pairs $(a_1, b_1), ..., (a_n, b_n)$ are found encoded in the sequence as we move form left to right in the dyadic notation for z.

C_1 follows from (2) and (1), as, because f is the longest frame in z (and by (1), there exists some frame f at all), it obviously contains a string of 1's longer than any of the string of 1's in the ordered pairs. If it didn't, this longer string of 1's would be enclosed by 2, thus meaning that this would be a frame, contrary to the assumption that f was the longest frame in z. C_2 follows from (2). In particular, because at least one sequence of the form $ffafbff$ is in z, we know that a, b is an ordered pair in the sequence. If we have n occurences of ffa_ifb_iff, then the sequence encoded in z is $(a_1, b_1), ..., (a_n, b_n)$. If there was any other sequence, then C_3 would not be satisfied, as the sequence found by moving from left to right along z is, as we have shown, $(a_1, b_1), ..., (a_n, b_n)$, not any other sequence. Thus, the sequence $(a_1, b_1), ..., (a_n, b_n)$ is the one and only sequence encoded in z. □

Problem 8

We are attempting to find some Σ_0 relation $K_0(x, y, z)$ such that for any finite sequence of ordered pairs of natural numbers, there is a number z such that for all numbers x and y, the relation $K_0(x, y, z)$ holds iff (x, y) is one of the ordered pairs in S. Let us define the set S_1 to be the set of all ordered pairs of integers $(x + 1, y + 1)$ such that the ordered pair (x, y) is a member of S. So, we then define $K_0(x, y, z)$ to be the following Σ_0 relation:
(1) $(\exists n \leq z)(\exists m \leq z)(n = x + 1 \wedge m = y + 1 \wedge K_1(m, n, z))$
And we know that there is some $K_1(m, n, z)$ relation such that z is a code number for S' and $K_1(m, n, z)$ holds iff $(m, n) \in S$. Hence, (1) holds iff
$K_1(x + 1, y + 1, z)$ (for the z that makes $K_1(m, n, z)$ encode S) holds iff $(x + 1, y + 1) \in S_1$ iff $(x, y) \in S$, thus completing the proof.

Problem 9

Proof. We are to show that the realation $g(\overline{x}) = y$ is Σ_1, that is, there is some formula of the form $\exists z R(x_1, ..., x_n, z)$ that represents $g(\overline{x}) = y$ where the relation $R(x_1, ..., x_n, z)$ is Σ_0. Well, let us first consider the set S of a sequence of ordered pairs of natural numbers:
$(0, g(\overline{0})), (1, g(\overline{1})), ..., (n, g(\overline{n}))$
where $n = x$ in the equation $g(\overline{x}) = y$. To be clear, every $g(\overline{i})$ in S shall not be the Dyadic Gödelian representation that $g(\overline{i})$ is, but instead is the

decimal number that is designated by the *Dyadic Godelian representation for* i. Thus, every member of an ordered pair of S is a natural number. Now, let us recall the fact that, for each positive integer $i \leq x$, $g(\overline{i+1}) = g(\overline{i}) * 122$, and that $i = 0 \supset g(\overline{i}) = 12$. Now, what we wish to show next is that, if $g(\overline{x}) = y$, then there is some sequence of natural numbers such that:
(1) (m, n) is a member of S
(2) If some (a, b) is a member of S, then so is (c, d) for $(a, b) = (c + 1, d * 122)$ or $(a = 0, b = \overline{4})$.
Well, by definition, every $g(\overline{x}) = y$ means that either $x = 0$, $y = \overline{4}$, or that, for some number a, b, $(x, y) = (a + 1, b * 122)$, and similarly for (a, b), and so on. Thus S contains this sequence. It can be seen by induction on every x that $g(\overline{x}) = y$ holds for every (x, y) in S, and thus, if we have some sequence S according to these conditions, it follows that $g(\overline{m}) = n$ holds. Hence, we have the following Σ_1 formula expressing $g(\overline{x}) = y$:
$\exists z[K_0(x, y, z) \wedge (\forall a \leq z)(\forall b \leq z)(K_0(a, b, z) \supset$
$[(a = 0 \wedge b = \overline{4})$
$\vee (\exists c \leq z)(\exists d \leq z)(K_0(c, d, z) \wedge a = c + 1 \wedge b = d * 122)])]$
Now let's be clear as to why this formula works. Assume that the above formula holds. Thus there is some z such that z is a code number for at least $(x + 1, y + 1)$, where the set S_1 it is a code number for has the corresponding set S (defined in Problem 8) which clearly satisfies conditions 1, 2, and thus $g(\overline{x}) = y$. On the other hand, assume that $g(\overline{x}) = y$. Thus, there is *some* number z such that z enocdes the set S_1 which corresponds to the set S which is one of the sets satisfying conditions 1, 2 for (x, y), thus meaning the formula holds. *Note*: We very well could have instead written $K_0(x, y, z)$ as $(\exists n \leq z)(\exists m \leq z)(n = x + 1 \wedge m = y + 1 \wedge K_1(m, n, z))$, but we wrote it as we did for simplicity. □

Exercise 2

We are to show that the relation $x^y = z$ is Σ_1.

Proof. Consider a set S of a sequence of ordered pairs such that:
(1): $(y, z) \in S$
(2): If $(a, b) \in S$ than either $a = 0 \wedge b = 1$ or there is some $(c, d) \in S$ such that $(a, b) = (c + 1, d \times x)$
where x, y, z are defined from the equation $x^y = z$. On the one hand, assume there is a set S satisfying conditions (1) and (2). It can easily be shown with generalized induction, where the component of (a, b) is $(a - 1, b/x)$ such that $(0, 1)$ has no components, that, for every $(a, b) \in S$, $x^a = b$, and thus, in particular, $x^y = z$. Conversley, assume that $x^y = z$. Hence we place $(y, z) \in S$, and then it is obvious that we can satisfy condition (2), as, by definition of a power, either $y = 0 \wedge z = 1$ or we can place the ordered pair (a, b) such that

$(y, z) = (a + 1, b \times x)$ (we can do this $y - 1$ times, as a power x^y is defined to have y factors of x). Thus $x^y = z$ iff there is a set S satisfying conditions $(1), (2)$. Hence, using Lemma K_0, we receive the following formula Σ_1 formula:
$\exists z[K_0(c, d, z) \wedge (\forall a \leq z)(\forall b \leq z)(K_0(a, b, z) \supset$
$[(a = 0 \wedge b = 1) \vee (\exists m \leq z)(\exists n \leq z)(a = m + 1 \wedge b = d \times e)])]$
that expresses the relation $x^y = z$. For, assume the formula holds. Hence, there is some z such that z is a code number for a sequence of natural numbers in a set S such that the set S_1 obviously satisfies conditions $(1), (2)$. Conversley, if $x^y = z$, then there is some sequence of pairs of natural numbers satisfying conditions $(1), (2)$, and the corresponding set S_1 can be encoded by *some* code number z. □

Problem 10

We consider some Σ_0 relation $R(x, y, z_1, ..., z_n)$ and consider the relation $\exists x \exists y R(x, y, z_1, ..., z_n)$. We are to show that this is Σ_1. Well, the above is equivalent to the following Σ_1 relation:
$\exists a[(\exists x \leq a)(\exists y \leq a)R(x, y, z_1, ..., z_n)]$
For, assume the above formula holds. This obviously implies that the initial relation holds. Conversley, assume that the initial formula holds. Then, there is some a such that the above Σ_1 relation holds, namely some a such that $a \geq x \wedge a \geq y$. Then, assume the relation $R(x, z_1, ..., z_n)$ is Σ_1. By definition, it follows that it is equivalent to some relation $\exists y R_1(x, x_1, ..., x_n, y)$ where $R_1(x, x_1, ..., x_n, y)$ is Σ_0. Hence, by the above result, we have $\exists x R(x, z_1, ..., z_n)$ being equivalent to the Σ_1 relation $\exists x \exists y R_1(x, x_1, ..., x_n, y)$. (It should be noted that, for both of these problems, it is of course not a requirenment that the variables being quantified are the first and second. The proofs that these two relation are Σ_1 works for any free variable at all in the relation are similar , just by replacing x by x_i, and y by y_i, and so forth).

Problem 11

We are attempting to show that there is some Σ_1 function $r(x, y)$ such that for any e the Gödel number of any expression E, and any number n, $r(e, n)$ is the Gödel number for $E[\overline{n}]$. Well, $E[\overline{n}]$ is:
$\exists v_1(v_1 = \overline{n} \wedge F(\overline{n}))$
Well, we let $\exists v_1(v_1 =$ have the Gödel number g, $\wedge$ is g_8 while $F(\overline{n})$ has the Gödel number f, and) has the Gödel number g_4. So, the Gödel number of the above formula is $g * g(\overline{n}) * g_8 * f * g_4$. So, $r(x, y)$ is this formula. Now, we must show that $r(x, y) = z$ is Σ_1. Well, this is equivalent to:

$\exists w(w = g(\overline{n}) \wedge g * w * g_8 * f * g_4 = z)$
Recall that $w = g(\overline{n})$ was shown to be equivalent to some formula of the form $\exists z R(x, y, z)$ where $R(x, y, z)$ is Σ_0. Hence, we can rewrite the above formula as:
$\exists w \exists v(R(w, y, v) \wedge g * w * g_8 * f * g_4 = z)$
Then, because the realation $R(w, y, v) \wedge g * w * g_8 * f * g_4 = z$ is Σ_0, the above formula is Σ_1 by Problem 10.
Note to self : To be *very* clear (as I spent numerous hours highly puzzled over this), when Smullyan says that we have shown $r(x, y) = z$ to be arithmetic, he does *not* mean $r(x, y)$ to be any relation at all (as I wrongly thought); he of course means it to be the $r(x, y)$ defined in this problem.

Problem 12

Assume A is Σ_1. Hence there is a Σ_0 relation $R(v, w)$ such that $v \in A$ iff $\exists w R(v, w)$. Also, $r(x, x) = y$ has been shown to be Σ_1, and hence there is a Σ_0 relation $R(x, y, z)$ such that $r(x, x) = y$ holds iff $\exists z R(x, y, z)$ is true. Hence $r(x, x) = y \wedge y \in A$ iff $\exists z \exists w(R_1(x, y, z) \wedge R_2(y, w))$, and this is Σ_1 by Problem 10, thus meaning $r(x, x) = y \wedge y \in A$ is Σ_1, and hence:
$\exists y(r(x, x) = y \wedge y \in A)$, that is, the formula that expresses $A^{\#}$, is also Σ_1 (also by Problem 10).

Problem 13

We consider two Σ_1 sets A, B. Thus, A, B are expressed by the two formulas $\exists z R_1(x, z)$, $\exists y R_2(x, y)$, respectively. Hence, the Σ_1 formula $\exists z(\exists x \leq z)(\exists y \leq z)(R_1(v, x) \wedge R_2(v, y))$ clearly represents $A \bigcap B$, as it is true iff $v \in A \wedge v \in B$. Also, the Σ_1 formula $\exists z(\exists x \leq z)(\exists y \leq z)(R_1(v, x) \wedge R_2(v, y))$ represents $A \bigcup B$, as it is true iff $v \in A \vee v \in B$.

Problem 14

We have already shown that every arithmetic set has a Gödel sentence, so the below result holds true:
Well, we consider any set W_0 of Gödel numbers of true sentences that is arithmetic. Thus the complement $\widetilde{W_0}$ of W_0 is also arithmetic, and therefore has some Gödel sentence X. Hence, by definition of a Gödel sentence, X is true iff x^*, for x^* the Gödel number of X, is in $\widetilde{W_0}$ iff $x^* \notin W_0$. Hence, X must be true, as if it was not, $x^* \in W_0$ and X would be false, contrary to the assumption that only Gödel numbers of true sentences were members of W_0.